Project 2.2: LLM Travel Planner Using RLHF, PPO, DPO, and Q-Learning

Problem Statement: You are building an **LLM-based travel assistant** for students travelling from **Kolkata to IIT Bhubaneswar** to attend a **PW course meet**. The assistant should generate travel plans that are budget-friendly, safe, time-feasible, clear, student-friendly, and robust to delays.

User Request

"I need to travel from Kolkata to IIT Bhubaneswar for a PW course meet. I want a budget-friendly plan. I prefer train travel if possible. I should reach IIT Bhubaneswar before 9 AM. Avoid unsafe late-night waiting. Suggest a simple travel plan with backup options."

Tasks

1. Use a small LLM such as **Qwen2.5-0.5B-Instruct** or **SmolLM2-360M-Instruct** to generate at least **3 candidate travel plans** for the given request. (These models you can run using T4-gpu of google colab)

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2. Design a **reward function** to score each plan based on budget friendliness, reaching before 9 AM, train preference, safety, clarity, backup options, and avoiding vague or infeasible suggestions.

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3. Rank the generated plans using your reward function and explain how this simulates **human feedback in RLHF**.

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4. Implement a toy **PPO actor-critic simulation**, where **state = user travel request**, **action = selected travel plan**, **actor = chooses one plan**, **critic = predicts expected plan quality**, **reward = score from reward function**, and **advantage = actual reward – critic prediction**.

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5. Implement a simple **Q-learning formulation** for the travel-planning task. Define states such as current city/time/budget status, actions such as train/bus/cab/wait/backup option, rewards such as reaching before 9 AM, staying within budget, avoiding unsafe waiting, and penalties for delay, excessive cost, or unsafe layovers. Show how the Q-table is updated for at least a few steps.

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6. Create a small **DPO preference dataset** with three fields: **prompt**, **chosen plan**, and **rejected plan**. The chosen plan should be safer, clearer, and more constraint-aware than the rejected plan.

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7. Compare **RLHF, PPO, DPO, and Q-learning** for this travel-planning task. Explain which method is easiest to implement and which is most suitable for improving an LLM-based travel assistant.